

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through **pure**
evidence.

— PERSISTENT WORLD · FRESH SCIENTIST · EVIDENCE FIRST

Problem → Motivation → System → Experience → Evaluation → Insights



Can an agent carry a research task to completion?

Turn a task contract into a tested model, a reproducible implementation, and evidence another researcher can inspect.

Explore

Choose hypotheses, write code, and run experiments.

Scientific judgment

Recover

Handle failures and revise unsuccessful ideas.

Long-running work

Deliver

Retain a valid model and the evidence behind it.

Verifiable progress

Why Current Research Agents Fall Short

Three obstacles to sustained autonomous research.

01 CLOSED-WORLD WORKFLOW



Predefined tools and recovery paths cannot handle the unexpected.

Unexpected problems require developer intervention.

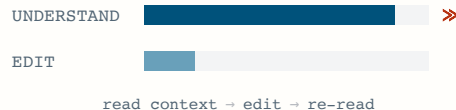
02 SINGLE TRAJECTORY



A long-lived context carries yesterday's momentum into today's search.

More iterations $\neq$ more exploration.

03 TREE SEARCH



Reconstructing file context costs far more tokens than the edit itself.

Context cost forces repeated reading.

Research is an **open-world** task. We need an **open-world** coding agent.

01 OPEN ACTION SPACE

-  Shell
-  Files
-  Browser
-  Git

Use what exists

02 SELF-RECOVERY

-  Read errors
-  Inspect docs
-  Inspect source
-  Install packages

Fix what breaks

03 RUNTIME EXTENSION

-  MCP
-  Skills
-  CLI
-  Packages

Add what's missing

WORKFLOW EXAMPLE · LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

Preserve the research world. Reset the researcher.

Two scientific roles. One safe runtime.

Scientist

Hypothesize · code · test · interpret

One trajectory

META

Audit · preserve · hand off

Across trajectories

Runtime

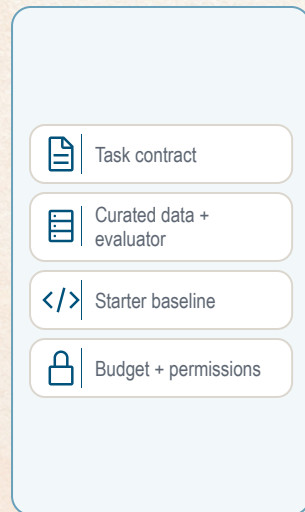
Isolation · cancellation · invariants

System lifetime

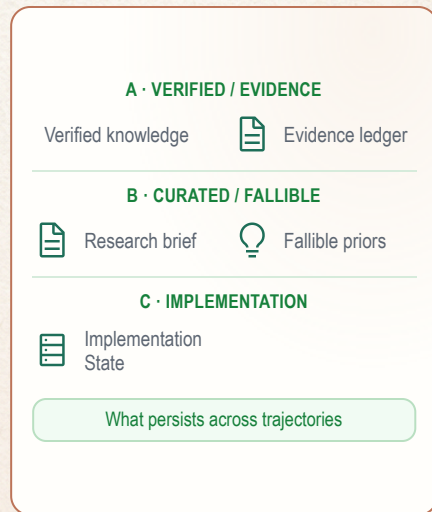
The Scientist chooses the science. META maintains the world. Runtime enforces mechanics, not scientific judgment.

The persistence boundary is explicit.

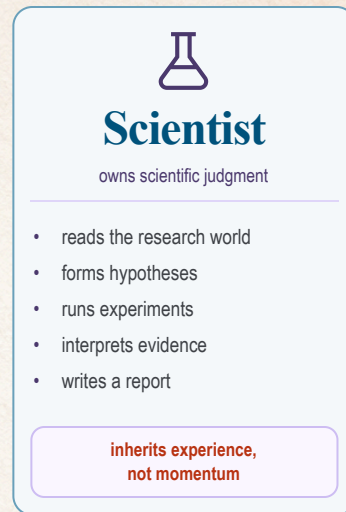
ENVIRONMENT



RESEARCH WORLD



FRESH SCIENTIST



META REVIEW



Selective persistence / trajectory handoff

audit → scope → preserve

Runtime — deterministic mechanics



Workspace isolation



Runner + evaluator



Logging

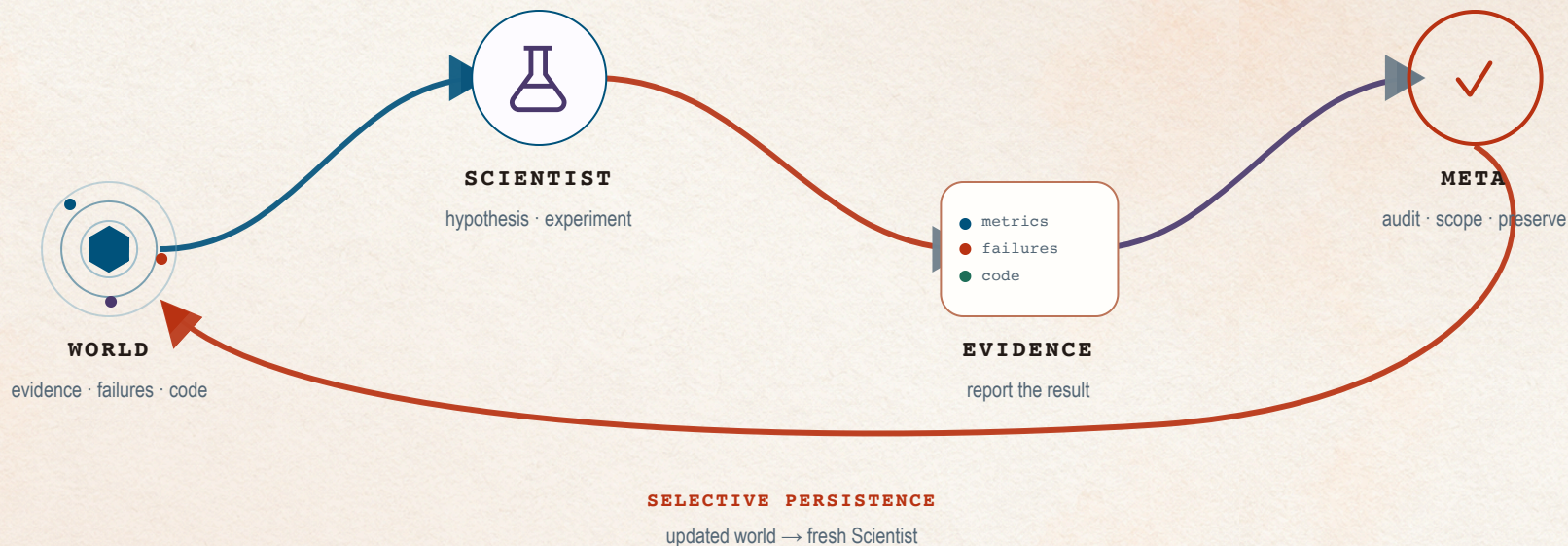


Artifact capture



State operations

Each cycle leaves a world to build on.



PERSISTS metrics · code · failures · original reports

RESTARTS an independent reasoning trajectory

Keep facts distinct from intuition.

research world

- ├ evidence & provenance
- ├ verified knowledge
- ├ current research state
- ├ fallible priors
- └ State + implementation

Built to be re-examined.

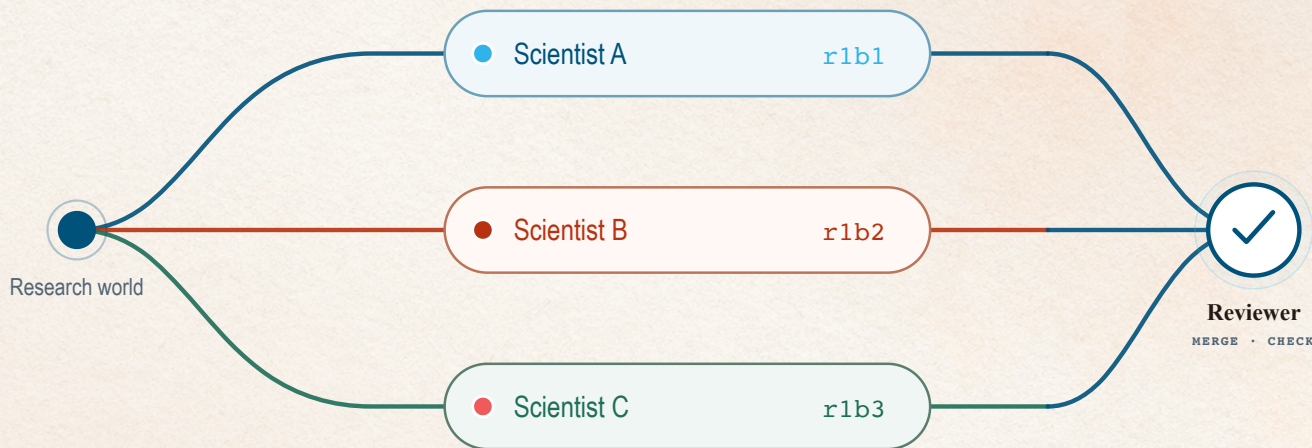
Facts have evidence.

Priors can be overturned.

Code can be picked up again.

Original reports remain available for audit.

One starting point. Independent answers.



Explore in isolated worktrees. Review after completion.

Optional synthesis: one implementation parent + reference evidence from other branches.
Adoption is explicit via parallel-promote. Code and memory are not merged automatically.

Start with the task and working constraints.

The operator defines the objective and constraints; the agent chooses and executes the experiments.

01 task.md

What to solve.

Objectives · constraints · evaluation

EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure · long_view · GAUC + nDCG@5

02 PERSONAL.md

How you want it solved.

Preferences · research style · priorities

EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment · never print credentials

Autonomous by default. Supervisable by design.

Evidence changed the next experiment.

E001–E013: reject weak ideas, improve representations, then test ensembles.

E001

Screen historical target encodings; reject weak variants.

Medium screening

E003

Establish a valid rich-FM checkpoint.

Full · 0.6016310

E008

Retain a five-seed, 38-field FM ensemble.

Full · 0.6040901

E013

Select the eight-seed, 46-field FM ensemble.

Full · 0.6059363

These are milestones within four autonomous cycles. Medium screens ideas; Full evaluations support retained score claims.

An evaluation failure became a recoverable step.

Cycle 1 completed training and evaluation, then failed while writing the evidence file.

Failure

NumPy float32 values broke JSON serialization.

Evidence writer

Repair

The Scientist converted scalars and reran.

Agent-authored fix

Retain

Printed measurements and saved evidence survived.

Auditable recovery

A retained state keeps the evidence inspectable.

Research Dashboard

RECORDED VALIDATION

★ Retained Validation Result

S004 · RETAINED STATE

SCORE-BEARING CHECKPOINT

S004

8-seed ensemble of 46-field Factorization Machine rankers incorporating leak-free historical user preferences (top tag, top tab, long-view rate bucket, interaction volume bucket, real-time tag/tab matches), secondary video tags, upload age, and per-seed Medium early stopping

retained checkpoint

GAUC	nDCG@5	Primary	Delta vs baseline
0.672842	0.539030	0.605936	+0.004336

Evidence · system/evidence/cycle4-fm-rich46-ensemble8-full.json

Research Intuition

Recent Activity

Run Summary

Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

METRIC	Official FM	Ours
GAUC	0.6674000	0.6728421
nDCG@5	0.5357000	0.5390304
Primary	0.6016000	0.6059363

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU

An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001–E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

Resource use is part of the evidence.

Project-level subscription estimate, with run-level telemetry reported separately.

~\$10

from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

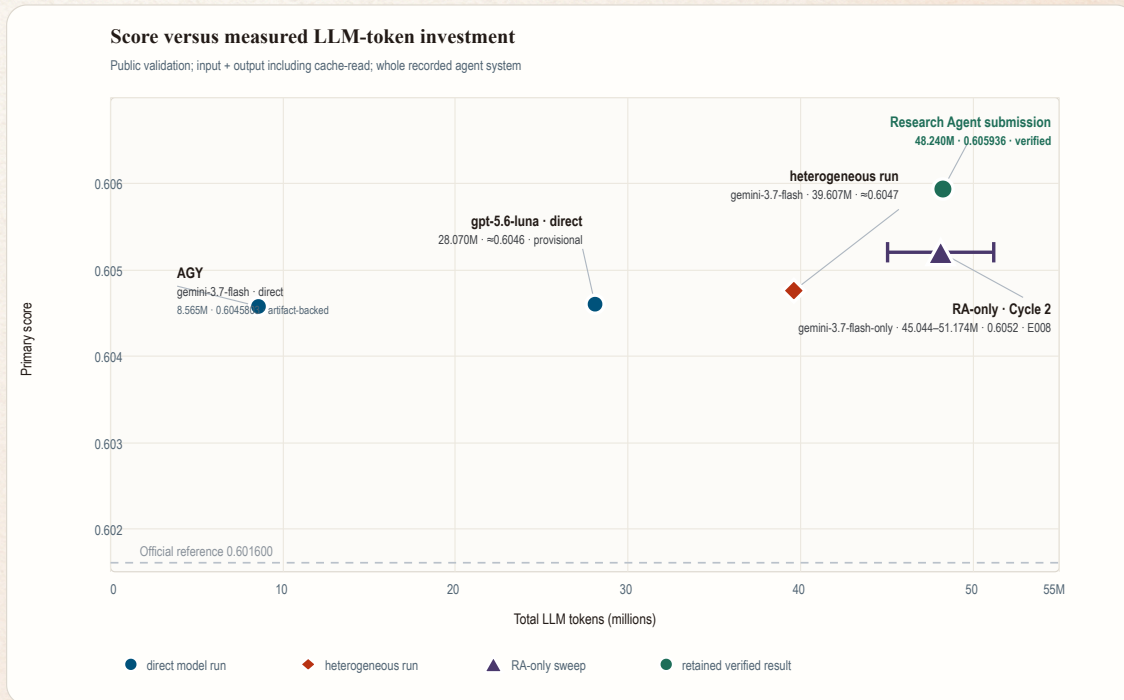
COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

The recorded runs used CPU training; LLM usage remains a separate cost.

Compare scores alongside their evidence.



Runs differ in model, budget, and evidence quality; this is not a controlled causal comparison.

The run taught us more than the final score.

What survived the experiments — and what remains unsettled.

01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

03 Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

02 A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

04 Reset $\neq$ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary 0.6055536; the canonical E001–E013 frontier remains 0.6059363.

4 cycles

0 post-launch interventions

0 GPU-hours

Reproducible evidence. Open questions.

01

One task does not establish generality.

External validity

02

META, State, and reset are not isolated.

Causal attribution

03

Summaries can lose context and anchor.

Memory quality

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS



Give every agent a branch.

main

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP



Leave the next move in GitHub.

01

Issue / prompt

inject the task

02

GitHub Action

run the agent

03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Keep the experience. Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
  --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles · evidence
review



Shilin Xu

DATA / METRICS

Data workflow · contracts ·
alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing · integration
support



Jiran Li

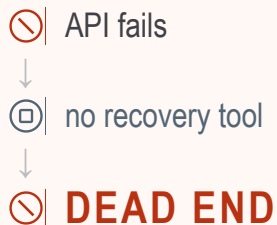
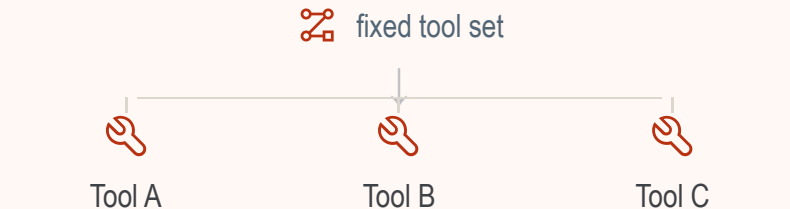
EVIDENCE /
REPRODUCIBILITY

Telemetry · documentation ·
packaging

The difference is not orchestration. It is recoverability.

LANGGRAPH-STYLE ORCHESTRATION

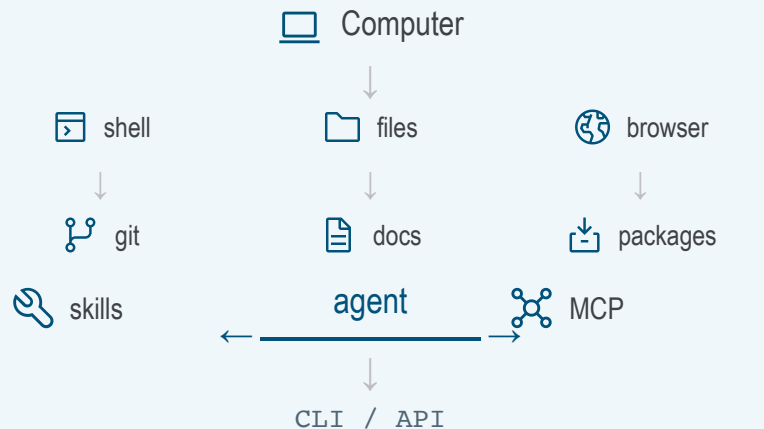
predefined graph



Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.